Representing Objects in Video as Space-Time Volumes by Combining Top-Down and Bottom-Up Processes

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Background

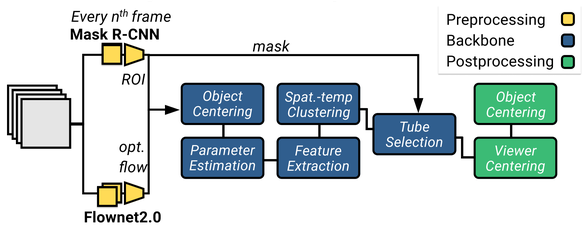
- Object Detection & unsupervised part detection in video
- Scalable spatio-temporal object representation
- Found structures correspond to temporally coherent object parts



Our approach bridges the conceptual gap between low-level grouping and instance segmentation representations.

Method

- Feature extraction from optical flow and object appearance
- Spatio-temporal scale selection, parameter estimation and clustering
- Viewer/Object-centered representation



Contributions

- Outperforms SoTA w.r.t. object part segmentation at metrics focusing on temporal coherence
- Introduces metrics for the evaluation of object decomposition in video
- Extends the DAVIS2016 dataset to include densely annotated object parts

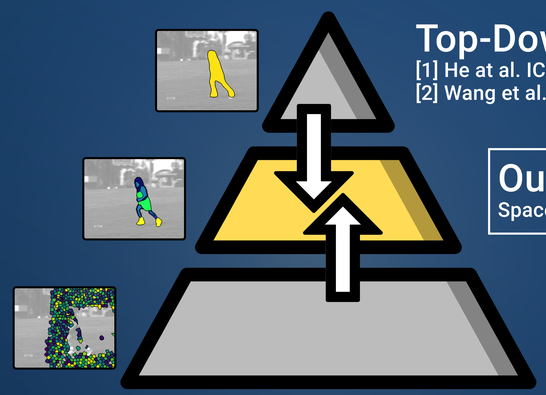
We extract temporally consistent **object parts** by **clustering features** extracted from **motion and appearance**.

Keywords

- Video Object Representation
- Temporally Coherent Segmentation
- Unsupervised Decomposition

Possible Applications

- Analyze motion patterns
- Visualize motion trajectories
- Extract features for action recognition
- Represent interactions between multiple objects



Top-Down Recognition
[1] He at al. ICCV 2017
[2] Wang et al. CVPR 2019

Our Approach
Space-Time Volumes of Interest

Bottom-Up Grouping
[3] Achanta et al. PAMI 2012
[4] Chang et al. CVPR 2013

Take a picture to view the **project page**
f-ilic.github.io/STVI

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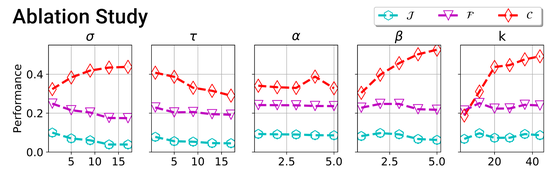
Results & Ablation Study

Object part segmentation results on DAVIS Parts

	$J\uparrow$	$F\uparrow$	$C\downarrow$	M
Mask R-CNN [1]	0.05	0.20	-	-
Siam Mask [2]	0.02	0.17	-	-
SLICO [3]	0.04	0.18	0.57	16.2
TSP [4]	0.06	0.16	0.49	78.8
Ours	**0.16**	**0.24**	**0.21**	3.7

J: Region Similarity
F: Contour Accuracy
C: Label Inconsistency
M: Average # of Tubes

Ablation Study

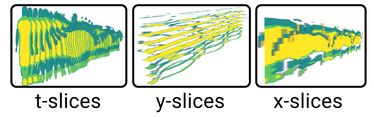


Our method is capable of estimating all necessary parameteres from the input video. Our expectations of a well behaved spatio-temporal representation w.r.t. parameter variations are met.

Visualizations

Representations can be extracted at different scales. This supports application specific control, both spatially and temporally.





Slicing along any STVI dimension can reveal interesting patters, e.g. the weaving of the legs.

Exemplary frames from DAVIS 2016 and the corresponding Space-Time Volumes



Method Details

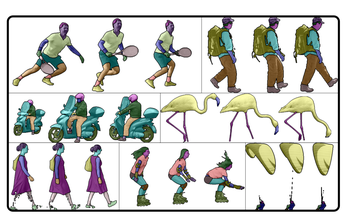
The 8-dimensional descriptor used for SLIC clustering is built by including optical flow, centered and normalized w.r.t. the object, and the object's appearence.

$$\phi(x,y,t) = \begin{bmatrix} \alpha(x,y,t)^T \\ (L,a,b)^T \\ \beta(\hat{F}_u,\hat{F}_v)^T \end{bmatrix}_{8\times 1}$$

$$L(\cdot;\sigma^2,\tau^2) = g(\cdot;\sigma^2,\tau^2) * f(\cdot)$$

We use spatio-temporal Gaussian Scale Space to select the scale at which the object representation is extracted.

DAVIS Parts dataset



We extend the popular DAVIS 2016 dataset with part annotations. The dataset is densely annotated with pixel accuracy and contains a variety of different object categories.